Hyperbolic Functions: Osborn's Rule & Proof

Questions 1–57

Question 1. By considering Osborn's Rule, or otherwise, write down the analogous hyperbolic identity for each of these trigonometric identities:

a) $\sin^2 x + \cos^2 x \equiv 1$
b) $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
c) $\tan^2 x + 1 \equiv \sec^2 x$
d) $\sin(x + y) \equiv \sin x \cos y + \cos x \sin y$
e) $\sin(x - y) \equiv \sin x \cos y - \cos x \sin y$
f) $\cos(x + y) \equiv \cos x \cos y - \sin x \sin y$
g) $\cos(x - y) \equiv \cos x \cos y + \sin x \sin y$
h) $\tan(x + y) \equiv \dfrac{\tan x + \tan y}{1 - \tan x \tan y}$
i) $\tan(x - y) \equiv \dfrac{\tan x - \tan y}{1 + \tan x \tan y}$
j) $\sin 2x \equiv 2 \sin x \cos x$
k) $\cos 2x \equiv \cos^2 x - \sin^2 x$
l) $\cos 2x \equiv 2 \cos^2 x - 1$
m) $\cos 2x \equiv 1 - 2 \sin^2 x$
n) $\tan 2x \equiv \dfrac{2 \tan x}{1 - \tan^2 x}$
o) $\sin 3x \equiv 3 \sin x - 4 \sin^3 x$
p) $\cos 3x \equiv 4 \cos^3 x - 3 \cos x$
q) $\tan 3x \equiv \dfrac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$
r) $\sin 4x \equiv \cos x(4 \sin x - 8 \sin^3 x)$
s) $\cos 4x \equiv 8 \cos^4 x - 8 \cos^2 x + 1$
t) $\tan 4x \equiv \dfrac{4 \tan x - 4 \tan^3 x}{1 - 6 \tan^2 x + \tan^4 x}$

Question 2. Write the following hyperbolic expressions in terms of $\tanh x$, $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$ only:

a) $\dfrac{\sinh x}{\cosh^2 x}$
b) $\dfrac{\cosh x}{\sinh^2 x}$

Question 3. Simplify the following hyperbolic expressions as far as possible:

a) $\operatorname{cosech} x \sinh x \cosh x$

b) $\tanh x \operatorname{sech} x \cosh x$

c) $\tanh x \operatorname{cosech} x \coth x \sinh x$

Question 4. Simplify the following hyperbolic expressions, removing any quotients where possible:

a) $\dfrac{\operatorname{sech} x}{\operatorname{cosech} x}$

b) $\dfrac{\coth x}{\operatorname{sech} x}$

c) $\dfrac{\operatorname{cosech} x}{\tanh x}$

d) $\dfrac{\operatorname{sech} x}{\coth x}$

e) $\dfrac{\operatorname{sech}^2 x}{\coth x}$

f) $\dfrac{\operatorname{cosech}^2 x \coth^2 x}{\operatorname{sech} x \sinh x}$

Question 5. Given that k is a constant where $k > 1$, show that

$$-\ln(k + \sqrt{k^2 - 1}) = \ln(k - \sqrt{k^2 - 1}).$$

Question 6. Given that k is a constant, show that

$$-\ln(k + \sqrt{k^2 + 1}) \neq \ln(k - \sqrt{k^2 + 1}).$$

Question 7. Show that the equation $\cosh x = k$, where k is a constant such that $k > 1$, has two real solutions: $x = \ln(k + \sqrt{k^2 - 1})$ and $x = \ln(k - \sqrt{k^2 - 1})$.

Question 8. It is given that $\cosh x = 4$. Show that $x = \ln(4 \pm \sqrt{15})$ and hence $x = \pm \ln(4 + \sqrt{15})$.

Question 9. Express each of these expressions in terms of $\cosh x$:

a) $8\cosh^2 x + 7\sinh^2 x$

b) $3\sinh^2 x - 13\cosh^2 x$

Question 10. Solve $3\cosh^2 x - \sinh^2 x = 5$, writing your answers in exact form.

Question 11. Solve each of these equations, writing your answers in exact form:

a) $\cosh^2 x - \sinh x - 7 = 0$
b) $\sinh^2 x - 2\cosh x - 7 = 0$
c) $\cosh^2 x - 14\sinh x + 47 = 0$
d) $2\sinh^2 x - 13\cosh x + 8 = 0$
e) $3\cosh^2 x + 22\sinh x - 19 = 0$
f) $4\sinh^2 x - 4\cosh x - 11 = 0$

Question 12. Solve each of these equations, writing your answers in exact form:

a) $25\cosh^4 x - 40\cosh^2 x - 9 = 0$
b) $36\cosh^4 x - 36\sinh^2 x - 91 = 0$
c) $192\sinh^4 x - 172\cosh^2 x + 67 = 0$

Question 13. Starting from $\cosh^2 x - \sinh^2 x \equiv 1$:

a) derive an identity for $\coth x$ and $\operatorname{cosech} x$

b) derive an identity for $\tanh x$ and $\operatorname{sech} x$

Question 14. Solve each of these equations, writing your answers in exact form:

a) $2\operatorname{sech}^2 x - 3\tanh x = 0$

b) $3\tanh^2 x - 5\operatorname{sech} x - 1 = 0$

c) $\operatorname{cosech}^2 x - 2\coth x - 7 = 0$

d) $4\coth^2 x - 25\operatorname{cosech} x + 2 = 0$

Question 15. Solve each of these equations, writing your answers in exact form:

a) $8\operatorname{sech}^2 x - 6\tanh x + 1 = 0$

b) $10\tanh^2 x - 21\operatorname{sech} x + 3 = 0$

c) $24\operatorname{cosech}^2 x + 58\coth x + 59 = 0$

d) $25\coth^2 x - 73\operatorname{cosech} x = 31$

Question 16. Prove there are no real solutions to the equation $6\tanh^2 x + 17\operatorname{sech} x - 18 = 0$.

Question 17. Solve each of these equations, writing your answers in exact form:

a) $\sinh x = \operatorname{sech} x$

b) $\tanh x = \operatorname{sech} x$

c) $\sinh x = 2\coth x$

d) $3\cosh x = \coth x$

Question 18.

a) Sketch the graphs of $y = 2\cosh x$ and $y = \operatorname{cosech} x$ on the same axes.

b) Determine the exact coordinates of where the two graphs intersect.

Question 19. It is given that $\sinh x = 3$. Find the exact values of:

a) $\cosh x$
b) $\tanh x$
c) $\coth x$
d) $\operatorname{cosech} x$
e) $\operatorname{sech} x$

Question 20. It is given that $\operatorname{sech} x = \dfrac{1}{\sqrt{2}}$. Find the exact values of:

a) $\tanh x$
b) $\cosh x$
c) $\sinh x$
d) $\operatorname{cosech} x$
e) $\coth x$

Question 21. It is given that $48\coth^2 x + 34\operatorname{cosech} x - 53 = 0$ and x is positive. Find the exact values of:

a) $\operatorname{cosech} x$
b) $\sinh x$
c) $\cosh x$
d) $\operatorname{sech} x$
e) $\tanh x$
f) $\coth x$

Question 22. Simplify the following hyperbolic expressions:

a) $\dfrac{\operatorname{sech}^2 x}{1 - \tanh^2 x}$
b) $\dfrac{\coth^2 x - 1}{\operatorname{cosech}^2 x}$
c) $\dfrac{1 - \tanh^2 x}{\coth^2 x - 1}$
d) $\dfrac{3 - 3\coth^2 x}{8 - 8\tanh^2 x}$

Question 23. It is given that $\sinh(1+\ln 6) = \dfrac{36e^2-1}{12e}$.

a) Hence find the exact value of $\cosh(1+\ln 6)$ in terms of e.

b) Using your answer to (a), show that $\tanh(1+\sqrt{6}) = \dfrac{36e^2-1}{36e^2+1}$.

Question 24. It is given that $p(x) = 12x^3 - 7x^2 - 42x + 40$.

a) Show that $(4x-5)$ is a factor of $p(x)$.

b) Write $p(x)$ as a product of three linear factors.

c) Hence prove that there are no real solutions to the equation

$$12\operatorname{sech}^3\theta + 7\tanh^2\theta - 42\operatorname{sech}\theta + 33 = 0.$$

Question 25.

a) Find $\sinh(\ln 3)$ and $\cosh(\ln 3)$.

b) Find $\sinh(\ln 4)$ and $\cosh(\ln 4)$.

c) Using your answers to (a) and (b), find the exact value of $\sinh(\ln 4 - \ln 3)$.

Question 26.

a) Find $\sinh(\ln 5)$ and $\cosh(\ln 5)$.

b) Find $\sinh(\ln 6)$ and $\cosh(\ln 6)$.

c) Using your answers to (a) and (b), find the exact value of $\cosh(\ln 6 - \ln 5)$.

Question 27.

a) Find $\tanh(\ln 2)$.

b) Find $\tanh(\ln 7)$.

c) Using your answers to (a) and (b), find the exact value of $\tanh(\ln 2 + \ln 7)$.

Question 28. Write the following in the form $\sinh k$, where k is a constant:

a) $\sinh 3 \cosh 6 - \cosh 3 \sinh 6$

b) $\sinh 4 \cosh 5 + \cosh 4 \sinh 5$

Question 29. Write the following in the form $\cosh k$, where k is a constant:

a) $\cosh 3 \cosh 7 - \sinh 3 \sinh 7$

b) $\cosh 8 \cosh 9 + \sinh 8 \sinh 9$

Question 30. It is given that $\cosh x = 2$ and $\cosh y = 3$, where x and y are real. Find the exact value of:

a) $\cosh(x - y)$

b) $\sinh(x - y)$

Question 31. Simplify the following expressions:

a) $\sinh(x + \ln 2) + \cosh(x - \ln 2)$

b) $\sinh(x - \ln 3) - \cosh(x + \ln 3)$

c) $\cosh(x - \ln 4) - \sinh(x - \ln 4)$

Question 32. In each case, find the exact value that solves the equation:

a) $\sinh(x + \ln 2) = \cosh x$

b) $\cosh(x - \ln 3) = \cosh x$

c) $\cosh(x - \ln 4) = \sinh x$

Question 33. In each case, find the exact value that solves the equation:

a) $\sinh(x + \ln 2) = \cosh(x - \ln 4)$

b) $\cosh(x - \ln 9) = \sinh(x + \ln 3)$

c) $\cosh(x + 3\ln 2) = \cosh(x - \ln 20)$

Question 34. Prove the following hyperbolic identities:

a) $\coth x + \cosh x \equiv \coth x(1 + \sinh x)$

b) $\coth x + \operatorname{sech} x \equiv \operatorname{cosech} x \operatorname{sech} x(\sinh x + \cosh^2 x)$

c) $\coth x - \tanh x \equiv \operatorname{cosech} x \operatorname{sech} x$

Question 35. Prove the following hyperbolic identities:

a) $\cosh x - \operatorname{sech} x \equiv \sinh x \tanh x$

b) $\operatorname{cosech} x + \sinh x \equiv \cosh x \coth x$

c) $\operatorname{cosech} x \operatorname{sech} x \equiv \tanh x(\coth^2 x - 1)$

d) $\dfrac{\operatorname{cosech} x(\sinh x \tanh x - 1)}{\sinh x + \cosh x} \equiv \tanh x - \operatorname{cosech} x$

Question 36. Simplify the following expressions:

a) $8 \sinh 2x \cosh 2x$

b) $2 \cosh^2 5x - 1$

c) $1 + 2 \sinh^2 3x$

d) $9 \cosh^2 \frac{x}{4} + 9 \sinh^2 \frac{x}{4}$

Question 37. Solve the following equations, giving your answers in an exact form:

a) $\sinh 2\theta - 2\sinh\theta = 0$

b) $\sinh 2\theta = 3\sinh\theta$

c) $5\sinh 2\theta = 6\cosh\theta$

d) $8\cosh\theta - 3\sinh 2\theta = 0$

Question 38. Solve the following equations, giving your answers in an exact form:

a) $2\cosh 2\theta = 7\cosh\theta$

b) $3\cosh 2\theta + 11\sinh\theta = 0$

c) $3\sinh\theta - \cosh 2\theta = 0$

d) $\cosh 2\theta - 4\cosh\theta = 0$

Question 39. Solve the following equations, giving your answers in an exact form:

a) $\cosh 2\theta + 2\sinh\theta - 5 = 0$

b) $5\cosh 2\theta - 3\sinh\theta = 5$

c) $8\cosh\theta - 3\cosh 2\theta = 3$

d) $5\cosh\theta - 12\cosh 2\theta + 17 = 0$

Question 40. Solve the following equations, giving your answers in an exact form:

a) $2\sinh 2\theta\,\operatorname{cosech}\theta = 5$

b) $5\sinh 2\theta\,\operatorname{sech}\theta = 7$

c) $\cosh 2\theta\,\operatorname{cosech}\theta = 3$

d) $2\cosh 2\theta\,\operatorname{sech}\theta = 7$

Question 41. Solve the following equations, giving your answers in an exact form:

a) $5\tanh 2x - 9\tanh x = 0$

b) $4\tanh x - 3\tanh 2x = 0$

Question 42. It is given that $\sinh x = \frac{4}{3}$. Find the exact value of $\sinh 2x$.

Question 43. It is given that $\sinh x = -\frac{1}{3}$. Find the exact value of $\cosh 2x$.

Question 44. It is given that $\cosh x = \frac{9}{4}$. Find the exact value of $\tanh 2x$.

Question 45.

a) Use hyperbolic identities, including $\sinh(A \pm B) \equiv \sinh A \cosh B \pm \cosh A \sinh B$, to write $\sinh 3\theta$ in terms of $\sinh \theta$.

It is given that $x = \sinh \theta$ and that $20x^3 + 15x - 12 = 0$.

b) Find the value of θ in the form $p \ln q$ where p is a rational number and q is a positive integer.

c) Hence find the real root of the equation $20x^3 + 15x - 12 = 0$, giving your answer in the form $\frac{1}{2}(5^m - 5^n)$, where m and n are rational numbers.

Question 46. Prove the following hyperbolic identities:

a) $\coth^2 x - \tanh^2 x \equiv \operatorname{cosech}^2 x + \operatorname{sech}^2 x$

b) $\sinh^2 x \tanh^2 x \equiv \sinh^2 x - \tanh^2 x$

c) $\operatorname{cosech}^2 x - \operatorname{sech}^2 x \equiv 4 \operatorname{cosech}^2 2x$

Question 47. Prove the following hyperbolic identities:

a) $(\operatorname{sech} x - \tanh x)^2 \equiv 1 - 2\operatorname{sech} x \tanh x$

b) $(\operatorname{sech} x - \coth x)^2 \equiv 2 - \tanh^2 x - 2\operatorname{cosech} x + \operatorname{cosech}^2 x$

c) $(\operatorname{sech} x - \operatorname{cosech} x)^2 \equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x(\cosh 2x - \sinh 2x)$

Question 48.

a) Prove the following hyperbolic identity:

$$\frac{1+\cosh x}{\sinh x} - \frac{\sinh x}{1+\cosh x} \equiv 2\operatorname{cosech} x.$$

b) Hence, where $t \neq 0$, solve the equation

$$\frac{1+\cosh t}{\sinh t} - \frac{\sinh t}{1+\cosh t} = -4.$$

Question 49.

a) Prove the following hyperbolic identity:

$$\frac{\sinh x}{\cosh x - 1} - \operatorname{cosech} x \equiv \coth x.$$

b) Hence, given $t \neq 0$, solve the equation

$$\frac{\sinh t}{\cosh t - 1} = 8 + \coth t.$$

Question 50. Show that the solution to the equation $\operatorname{sech} \frac{x}{2} = 6 \sinh \frac{x}{2}$ can be written in the form $\ln(p + \sqrt{q}) - \ln r$, where p, q and r are positive integers.

Question 51. Prove the following hyperbolic identities:

a) $\coth 2x - \tanh 2x \equiv 2\,\text{cosech}\, 4x$

b) $\coth x - \tanh 2x \equiv \coth x\,\text{sech}\, 2x$

Question 52. Prove the following hyperbolic identities:

a) $\coth 2x - \cosh 2x \equiv \coth 2x(1 - \sinh 2x)$

b) $\coth 2x + \text{cosech}\, 2x \equiv \coth x$

Question 53. Prove the following hyperbolic identity:

$$2\operatorname{cosech} 4x(\sinh^2 x + \cosh 2x) \equiv \tanh 2x + \operatorname{cosech} 2x.$$

Question 54. Prove the following hyperbolic identities:

a) $(\sinh 2x + \cosh 2x)^2 \equiv \sinh 4x + \cosh 4x$

b) $\dfrac{2\sinh x}{\cosh 2x - 1} - \dfrac{2\sinh x}{\cosh 2x + 1} \equiv \operatorname{cosech} x \operatorname{sech}^2 x$

Question 55. Prove the following hyperbolic identities:

a) $\cosh 2x - \sinh 2x \equiv \dfrac{1}{(\sinh x + \cosh x)^2}$

b) $\coth x(\cosh 2x - \sinh 2x) \equiv \dfrac{\cosh x(\coth x - 1)}{\sinh x + \cosh x}$

Question 56. Solve the equation $8\sinh x + 15\cosh x = 6\sinh x \cosh x + 20$, giving your answer in exact form.

Question 57. Solve the following equations:

a) $\sinh(\ln(8+2x)) = x$

b) $\cosh(\ln(3-5x)) + 10x = 0$

c) $\operatorname{sech}(\ln(4x+9)) + \frac{1}{12x} = 0$

d) $\tanh(\ln(\sqrt{3}+4x)) = 2x$

Hyperbolic Functions: Domain and Range

Questions 1–10

Question 1. For each of the following functions, sketch the graph and state whether it is one-to-one or if it is many-to-one.

a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

The answers for the following questions which require a range of values should be given in set notation.

Question 2. State the largest possible domain and range of the following functions:

a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

Question 3. State the range of the following, given $x < 0$:

a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

Question 4. State the range of the following, given $x \in \mathbb{R}$:

a) $f(x) = \sinh^2 x$
b) $y = \cosh x + 3$
c) $g(x) = \tanh(x - 2)$
d) $y = 2\sinh x$
e) $h(x) = -\cosh^2 x$
f) $y = \tanh^2 x + 5$

Question 5.

a) Write $2\sinh x + 4\cosh x$ in the form $ae^x + be^{-x}$ where a and b are real constants.

b) Hence explain why, for $x \in \mathbb{R}$, that $2\sinh x + 4\cosh x > e^{-x}$.

Question 6. Show clearly that, for $x \in \mathbb{R}$,

$$4\sinh^2 x - e^{-2x} > -2.$$

Prerequisite knowledge for Q7: Hyperbolic Double-Angle Formulae

Question 7.

a) Write $\sinh 2x$ in terms of $\sinh x$ and $\cosh x$.

b) Given $x > 0$, write down the range of:

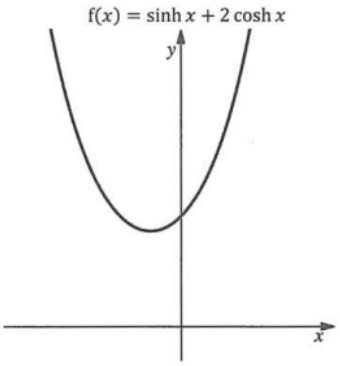
i) $y = \sinh x$

ii) $y = \cosh x$

c) Hence show that $\sinh 2x > \sinh x$ for $x > 0$.

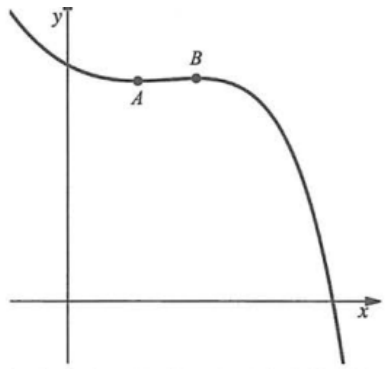
Prerequisite knowledge for Q8–Q10: Differentiation of Hyperbolic Functions

Question 8. The graph below shows $f(x) = \sinh x + 2\cosh x$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 9. The graph below shows the curve with equation $y = 3\cosh x - \sinh x \cosh x$. The points A and B on the curve are stationary points and have x-coordinates a and b respectively.



It is given that the function f is defined as $f(x) = 3\cosh x - \sinh x \cosh x$ where $x \in [a, b]$. Find the range of f in interval notation.

Question 10. It is given that

$$f(x) = \frac{e^{2x} + e^{-2x}}{(e^x + e^{-x})^2} - \tanh^2 x, \qquad x \in \mathbb{R}.$$

a) Show that $f(x) = 2(e^x + e^{-x})^{-2}$, $x \in \mathbb{R}$.
b) Find the coordinates of the stationary point of f.
c) State the range of f, giving your answer in set notation.

Hyperbolic Functions: Inverse Functions

Questions 1–33

Question 1. For each of the following functions, sketch the graph:

a) $y = \operatorname{arsinh} x$

b) $y = \operatorname{arcosh} x$

c) $y = \operatorname{artanh} x$

Question 2. Given that $x \in \mathbb{R}$, show that

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right).$$

Question 3. Given that $x \in \mathbb{R}$ and $k > 0$, show that

$$\operatorname{arsinh} \frac{x}{k} = \ln \left(\frac{x + \sqrt{x^2 + k^2}}{k} \right).$$

Question 4. Given that $x \geq 1$, show that

$$\operatorname{arcosh} x = \ln \left(x + \sqrt{x^2 - 1} \right).$$

Question 5. Given that $\frac{x}{k} \geq 1$ and $k > 0$, show that

$$\operatorname{arcosh} \frac{x}{k} = \ln\left(\frac{x + \sqrt{x^2 - k^2}}{k}\right).$$

Question 6. Given that $-1 < x < 1$, show that

$$\operatorname{artanh} x = \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right).$$

Question 7. Given that $-k < x < k$ and $k > 0$, show that

$$\operatorname{artanh} \frac{x}{k} = \frac{1}{2} \ln \left(\frac{k+x}{k-x} \right).$$

Question 8. Solve the equation $\operatorname{arcosh} x = \ln 10$, writing your answer in exact form.

Question 9. Solve the equation $\operatorname{arsinh} x = -\ln 6$, writing your answer in exact form.

Question 10. Solve the equation $\operatorname{artanh} \frac{x}{6} = -\ln 2$, writing your answer in exact form.

Question 11. Solve the following equations, writing your answers in exact form:

a) $\operatorname{artanh} x = \ln\left(\sqrt{x+2}\right)$

b) $\operatorname{artanh} \frac{x}{3} = \ln\left(\sqrt{3-2x}\right)$

c) $\operatorname{artanh} 5x = \ln\left(\sqrt{2+5x}\right)$

d) $4x + 1 = \tanh(\ln(\sqrt{5-2x}))$

Question 12. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arsinh} x$

b) $y = \operatorname{arcosh} x$

c) $y = \operatorname{artanh} x$

Question 13. State the exact range of the following functions, given $-\frac{1}{2} < x < \frac{1}{2}$:

a) $f(x) = \operatorname{arsinh} x$

b) $g(x) = \operatorname{artanh} x$

Question 14. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arcosh} x + 1$

b) $y = \operatorname{arcosh}(x - 1)$

c) $y = \operatorname{arcosh} 2x$

d) $y = 2 \operatorname{arcosh} x$

e) $y = 3 \operatorname{arcosh}(x + 2)$

f) $y = \operatorname{arcosh}(2x - 3)$

g) $y = 4 \operatorname{arcosh} x - 5$

h) $y = \operatorname{arcosh}(4 - x)$

i) $y = 8 - 3 \operatorname{arcosh} x$

j) $y = -\operatorname{arcosh}\left(9 - \frac{1}{2}x\right)$

Question 15. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arsinh}^2 x$

b) $y = \operatorname{arcosh}^2 x$

c) $y = \operatorname{artanh}^2 x$

Question 16. Given that $x \geq 1$, explain why $\operatorname{arsinh} x > \operatorname{arcosh} x$.

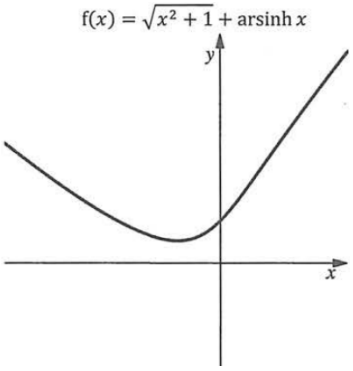
Question 17. Given that $x \geq 1$, state the exact range of $f(x) = \operatorname{arsinh} x + \operatorname{arcosh} x$.

Question 18. Solve the equation

$$\operatorname{artanh} x + \ln \sqrt{1-x} = \frac{1}{2} \ln 3 - 2 \ln 2.$$

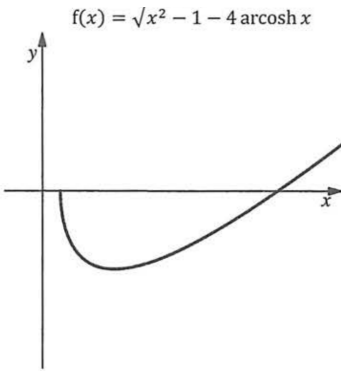
Question 19. Solve the equation $\operatorname{arsinh} x = \ln 4 - \ln 3 + \operatorname{arcosh} x$.

Question 20. The graph below shows $f(x) = \sqrt{x^2+1} + \operatorname{arsinh} x$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 21. The graph below shows $f(x) = \sqrt{x^2 - 1} - 4\,\text{arcosh}\,x$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 22. It is given that $f(x) = 2\,\text{arsinh}\,x - \text{arcosh}\,x$.

a) Explain why the largest possible domain of f is $\{x : x \in \mathbb{R},\ x \geq 1\}$.

b) Show that $y = f(x)$ has a single stationary point.

c) Show that the range of f is greater than or equal to

$$2\ln(\sqrt{15} + 2\sqrt{6}) - \ln(\sqrt{15} + \sqrt{6}) - \ln 3.$$

Question 23. Given that $x \neq 0$, show that

$$\operatorname{arcosech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right).$$

Question 24. Given that $x \neq 0$ and $k > 0$, show that

$$\operatorname{arcosech} \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right).$$

Question 25. Given that $0 < x < 1$, show that

$$\operatorname{arsech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right).$$

Question 26. Given that $0 < \frac{x}{k} < 1$ and $k > 0$, show that

$$\operatorname{arsech} \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 - x^2}}{x}\right).$$

Question 27. Given that $x < -1$ or $x > 1$, show that

$$\operatorname{arcoth} x = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right).$$

Question 28. Given that $x < -1$ or $x > 1$, and $k > 0$, show that

$$\operatorname{arcoth} \frac{x}{k} = \frac{1}{2} \ln \left(\frac{x+k}{x-k} \right).$$

Question 29. Solve the equation $\operatorname{arcosech} x = \ln 3$, writing your answer in exact form.

Question 30. Solve the equation $\operatorname{arsech} \frac{x}{4} = \ln 5$, writing your answer in exact form.

Question 31. Solve the equation $\operatorname{arcoth} x = \ln 2$, writing your answer in exact form.

Question 32. Solve the following equations, writing your answers in exact form:

a) $\operatorname{arcosech} x = \ln\left(2 + \frac{1}{x}\right)$

b) $\operatorname{arsech} \frac{x}{2} = \ln\left(3 - \frac{4}{x}\right)$

c) $\operatorname{arcoth} 3x = \ln\sqrt{3 + 4x}$

Question 33. Solve the equation $\operatorname{artanh} \frac{x}{2} = \operatorname{arcoth}(3x + 1)$.

Hyperbolic Functions: Reciprocal Functions

Questions 1–28

Question 1. Write down the exponential form of each of these, giving your answer:

a) $\operatorname{cosech} x$
b) $\operatorname{sech} x$
c) $\coth x$

Question 2. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\operatorname{cosech} 1$
b) $\operatorname{sech} 1$
c) $\operatorname{cosech}(-1)$
d) $\operatorname{sech}(-1)$

Question 3. Show that $\operatorname{cosech}(-x) = -\operatorname{cosech} x$.

Question 4. Show that $\operatorname{sech}(-x) = \operatorname{sech} x$.

Question 5. By using the exponential forms of $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$, find a simplified exact value for each of these:

a) $\operatorname{cosech}(\ln 3)$
b) $\operatorname{sech}(\ln 6)$
c) $\operatorname{cosech}(\ln \sqrt{5})$
d) $\operatorname{sech}(4 \ln 3)$

Question 6. Show that $\operatorname{cosech}(\ln k) = \dfrac{2k}{k^2 - 1}$ where k is a positive constant.

Question 7. Show that $\operatorname{sech}(3\ln k) = \dfrac{2k^3}{k^6+1}$ where k is a positive constant.

Question 8. It is given that $f(x) = \operatorname{cosech}(2\ln x) - \operatorname{sech}(2\ln x)$ where $x > 0$ and $x \neq k$ for some constant k.

a) Explain why $k = 1$.

b) Show that $f(x) = \dfrac{4x^2}{(x^4-1)(x^4+1)}$.

Question 9. By using the exponential forms of $\sinh x$ and $\cosh x$, show that $\coth x = \dfrac{e^{2x}+1}{(e^x+1)(e^x-1)}$.

Question 10. By using the exponential forms of $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$, find a simplified exact value for each of these:

a) $\coth(\ln 2)$
b) $\coth(\ln \sqrt{2})$
c) $\coth\left(\frac{1}{4}\ln 8\right)$

Question 11. Show that $\coth(-x) = -\coth x$.

Question 12. Find a simplified exact value for each of these:

a) $\coth(-\ln 3)$

b) $\coth(-5\ln 2)$

Question 13. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech} x = 1$

b) $\operatorname{cosech} x = 2$

c) $\operatorname{cosech} x = -\frac{1}{5}$

Question 14. Solve the following equations, writing your answers in exact form:

a) $\operatorname{sech} x = 1$

b) $\operatorname{sech} x = \frac{1}{2}$

c) $\operatorname{sech} x = \frac{2}{3}$

Question 15. Solve the following equations, writing your answers in exact form:

a) $\coth x = 2$

b) $\coth x = \frac{3}{2}$

c) $\coth x = -\frac{5}{3}$

Question 16. Solve the following equations, writing your answers in exact form:

a) $5\,\mathrm{sech}\, x - 2\tanh x = 5$

b) $8\,\mathrm{cosech}\, x - 3\coth x = 27$

c) $2\tanh x - 5\coth x = 9$

Question 17. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech}^2 x = 9$

b) $\operatorname{sech}^2 x = \frac{1}{16}$

c) $\coth^2 x = 3$

Question 18. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech}^2 x + \operatorname{cosech} x - 20 = 0$

b) $10 \operatorname{sech}^2 x - 9 \operatorname{sech} x + 2 = 0$

c) $3 \coth^2 x - 16 \coth x - 64 = 0$

Question 19. For each of the following functions, sketch the graph and state whether it is one-to-one or if it is many-to-one:

a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Use set notation for the following questions which require a range of values.

Question 20. State the largest possible domain and range of the following functions:

a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Question 21. State the range of the following functions given $x > 0$:

a) $y = \operatorname{cosech} x$

b) $y = \operatorname{sech} x$

c) $y = \coth x$

Question 22. Sketch each of the following functions, stating their largest possible domain and range:

a) $f(x) = \operatorname{cosech}(x + 3)$

b) $g(x) = 2 + \operatorname{sech} x$

c) $h(x) = \coth(x - 1) - 1$

Question 23. State the largest possible domain and range for each of the following functions:

a) $f(x) = \operatorname{cosech}(x-2)+2$
b) $y = 2 - \operatorname{cosech} x$
c) $g(x) = 3\operatorname{sech}(x+3)$
d) $y = 4 + 3\operatorname{sech} x$
e) $h(x) = \coth(x+4) - 3$
f) $y = 2 - \frac{1}{3}\coth x$

Question 24. State the largest possible domain and range for each of the following functions:

a) $y = \operatorname{cosech}^2 x$
b) $y = \operatorname{sech}^2 x$
c) $y = \coth^2 x$

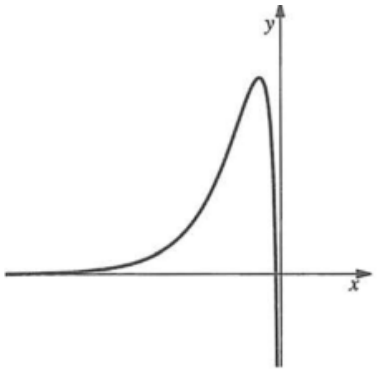
Question 25.

a) Write $\operatorname{cosech} x + \operatorname{sech} x$ in the form $\dfrac{ae^{bx}}{e^{cx} - 1}$ where a, b and c are real constants.

b) Hence explain why, for $x > 0$, $\operatorname{cosech} x + \operatorname{sech} x > \dfrac{a}{e^{cx} - 1}$.

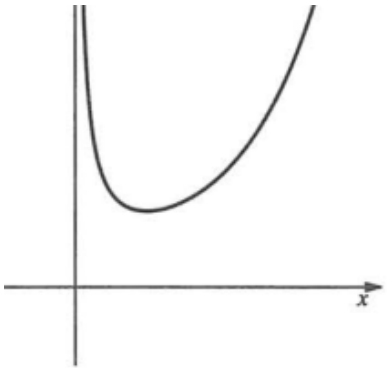
Question 26. Show clearly that, for $x > 0$, $\coth x - \tanh x > 4e^{-2x}$.

Question 27. The graph below shows $f(x) = \operatorname{cosech} x + 8 \operatorname{sech} x$ where $x < 0$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 28. The graph below shows $f(x) = \cosh x + \coth x$ where $x > 0$.



Find the exact largest possible range of f, writing your answer in interval notation.

Graph Sketching: Reciprocal Hyperbolic Functions

Questions 1–8

Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

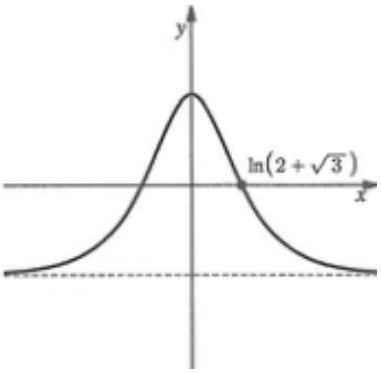
a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Question 2.

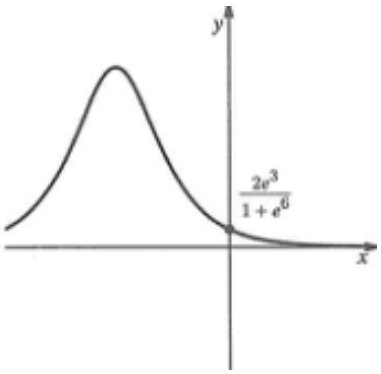
a) By sketching the curves with equations $y = \cosh x$ and $y = \operatorname{cosech} x$ on the same axes, show that they have exactly one point of intersection.
b) Find the x-coordinate of the point of intersection of the two curves, writing your answer in the form $\frac{1}{2}\ln(a+\sqrt{b})$, where a and b are positive integers.

Question 3. Find the equation of each of the following graphs in the form $y = \text{sech}(x - a)$ or $y = \text{sech}\, x + b$, where a and b are integers:

a)



b)



Question 4. State the equations of the asymptotes of the following graphs:

a) $y = \coth(x + 1) + 1$

b) $y = \coth(x - 2) - 3$

Question 5. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \coth x + 2$
b) $y = \coth(x - 3)$
c) $y = \coth(x - 1) + 1$

Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

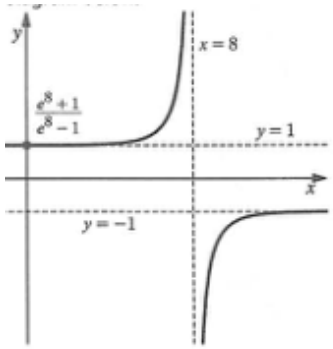
a) $y = 2\coth x$
b) $y = \frac{1}{3}\coth x$
c) $y = -\coth x$

Prerequisite knowledge for Q7–Q8: Combinations of Graph Transformations

Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 2 - \operatorname{cosech} x$
b) $y = -\frac{1}{3}\operatorname{sech} x$
c) $y = 2\coth x - 1$
d) $y = \operatorname{cosech}(2x + 1)$
e) $y = \operatorname{sech}(3 - x)$
f) $y = \coth\left(\frac{1}{2}x - 3\right)$

Question 8. Suggest an equation for the curve shown in the diagram below:



Hyperbolic Functions: Differentiation

Questions 1–49

Question 1. Show, by writing $\sinh x$, $\cosh x$ and $\tanh x$ in terms of exponentials, that:

a) $\dfrac{d}{dx}(\sinh x) = \cosh x$

b) $\dfrac{d}{dx}(\cosh x) = \sinh x$

c) $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

Prerequisite knowledge for Q2: Hyperbolic Identities

Question 2. It is given that $\lim\limits_{h\to 0} \dfrac{\sinh h}{h} = 1$ and $\lim\limits_{h\to 0} \dfrac{\cosh h - 1}{h} = 0$. Show, by first principles, that:

a) $\dfrac{d}{dx}(\sinh x) = \cosh x$

b) $\dfrac{d}{dx}(\cosh x) = \sinh x$

The following questions use the derivatives of $\sinh kx$, $\cosh kx$ and $\tanh kx$ and require no further Chain, Product or Quotient Rule.

Question 3. Differentiate each of the following:

a) $y = \sinh 2x$
b) $y = \cosh 3x$
c) $y = \tanh 6x$
d) $y = \sinh \frac{1}{4}x$
e) $y = \cosh \frac{x}{3}$
f) $y = \tanh\left(-\frac{x}{5}\right)$

Question 4. Differentiate each of the following:

a) $y = 3\sinh 4x - 6\tanh 3x$
b) $y = 5\cosh \frac{x}{10} + \frac{1}{2}\tanh \frac{x}{4}$

Question 5. Find the exact gradient of the curve $y = 3\tanh\frac{1}{2}x$ at $\left(\ln 16, \frac{45}{17}\right)$.

Question 6. Find the equation of the tangent to the curve $y = 3\sinh x$ when $x = \ln 6$, writing your answer in the form $ax + by = c$ where a, b and c are constants.

Question 7. Find the equation of the tangent to the curve $y = 5\sinh x - \cosh x$ when $x = \ln 2$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 8. Find the equation of the tangent to the curve $y = 3\cosh 2x - 4\sinh 3x$ when $x = \ln \frac{1}{3}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 9. Find the equation of the tangent to the curve $y = \tanh \frac{x}{2}$ when $x = \ln 4$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 10. Find the equation of the normal to the curve $y = 8 + 6\tanh \frac{1}{3}x - 4\tanh \frac{2}{3}x$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

Question 11. The line l_1 is the normal to the curve $y = 12\sinh\frac{x}{3}$ when $x = \ln 8$, and the line l_2 is the normal to the curve $y = 6\cosh\frac{x}{4}$ when $x = \ln 16$. Show that the lines l_1 and l_2 intersect when $x = a\ln 2 + b$, where a and b are rational numbers to be found.

Question 12. Show that the curve with equation $y = \sinh x + \tanh x$ is always increasing.

Question 13. It is given that the curve with equation $y = \sinh x + 2\cosh x$ has a single stationary point. Find the exact coordinates of the stationary point. **Prerequisite knowledge for Q14: Hyperbolic Double-Angle Formulae**

Question 14. Show that the curve with equation $y = 9\cosh x + 3\sinh 2x$ has no point where the gradient is 1.

Question 15. It is given that $f(x) = p\sinh x - q\cosh x$ where p and q are positive constants. Show that the curve $y = f(x)$ has no stationary points when $p \geq q$.

Question 16. It is given that the curve with equation $y = \sinh x - 8\tanh x$ has two stationary points. Find the exact coordinates of both stationary points.

Question 17. Show that the curve $y = 5\cosh x + \sinh x + x$ has a single stationary point.

Question 18. Determine whether the curve with equation $y = \cosh 3x - 2\sinh x$ is convex (concave upwards) or concave (concave downwards) when $x = \ln 2$.

Question 19. Show that the curve $y = 4\sinh 2x - 3\cosh 2x$ has a single point of inflection and find its exact coordinates. **The following questions use the derivatives of** $\sinh kx$**,** $\cosh kx$ **and** $\tanh kx$ **and use any differentiation knowledge from A-Level Maths.**

Question 20. Differentiate each of the following:

a) $y = \sinh(\sqrt{x})$

b) $y = \cosh\left(\frac{1}{\sqrt{x}}\right)$

c) $y = \tanh\left(\frac{x+1}{2\sqrt{x}}\right)$

Question 21. Differentiate each of the following:

a) $y = \sinh^2 x$
b) $y = \cosh^2 x$
c) $y = \sinh^3 x$
d) $y = \cosh^4 x$
e) $y = \tanh^2 x$
f) $y = \tanh^5 x$

Question 22. Show that the curve $y = 3\cosh^2 x - \sinh x$ has a single stationary point.

Question 23. It is given that $f(x) = \cosh^2 2x - p\cosh 2x$, where p is a constant. Find the minimum value of p such that $y = f(x)$ has more than one stationary point.

Question 24. Differentiate each of the following:

a) $y = \sinh(\cosh x)$

b) $y = \cosh(\sinh x)$

c) $y = \tanh(\tanh x)$

Question 25. Show that:

a) $\dfrac{d}{dx}(\ln(\sinh x)) = \coth x$

b) $\dfrac{d}{dx}(\ln(\cosh x)) = \tanh x$

Prerequisite knowledge for Q26: Hyperbolic Double-Angle Formulae

Question 26. Show that $\dfrac{d}{dx}\left(\dfrac{1}{2}\ln(\tanh x)\right) = \operatorname{cosech} 2x$.

Question 27. Differentiate each of the following:

a) $y = \sinh x \cosh x$

b) $y = \sinh 3x \cosh 4x$

c) $y = x \sinh x$

d) $y = x^2 \cosh x$

e) $y = (x+1) \sinh 2x$

f) $y = (1 - x^2) \cosh \frac{x}{2}$

Question 28. Find the exact gradient of the curve $y = \sinh 2x \tanh 3x$ at $\left(\ln 2, \frac{189}{104}\right)$.

Question 29. Find the equation of the tangent to the curve $y = (x^2 + x - 6)\cosh x$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are integers.

Question 30. Find the equation of the normal to the curve $y = \sinh(12x - \ln 2)\cosh(3x - \ln 3)$ when $x = 0$, writing your answer in the form $ax + by = c$ where a, b and c are integers.

Question 31. Show that the curve with equation $y = \sinh x \cosh x$ is always increasing.

Question 32. It is given that the curve with equation $y = (\cosh x - 3)\cosh x$ has three stationary points. Find the exact coordinates of all three stationary points.

Question 33. Determine whether the curve with equation $y = \sinh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right)$ is convex (concave upwards) or concave (concave downwards) when $x = \ln 64$.

Question 34. Show that the curve $y = (\sinh x - 6)\sinh x$ has two points of inflection and find their exact coordinates.

Question 35. By writing $\tanh x$ as $\dfrac{\sinh x}{\cosh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$.

Question 36. Differentiate each of the following:

a) $y = \dfrac{\sinh x}{\cosh 2x}$

b) $y = \dfrac{\cosh x}{\sinh 2x}$

c) $y = \dfrac{x}{\sinh x}$

d) $y = \dfrac{\cosh x}{x}$

e) $y = \dfrac{x+1}{\sinh 2x}$

f) $y = \dfrac{\cosh 3x}{x^2+1}$

Question 37. Find the exact gradient of the curve $y = \dfrac{\sinh 3x}{\sinh x}$ at $\left(\ln 2, \frac{21}{4}\right)$.

Question 38. Find the equation of the tangent to the curve $y = \dfrac{x + \cosh 3x}{1 - \sinh 2x}$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are integers.

Question 39. Find the equation of the normal to the curve $y = \dfrac{\tanh 3x + 6}{\cosh 2x + 3}$ when $x = 0$, writing your answer in the form $ax + by = c$ where a, b and c are integers.

Question 40. It is given that $f(x) = \dfrac{x+1}{\sinh x}$.

a) Show that the x-coordinates of the stationary points of the curve $y = f(x)$ satisfy $\tanh x = x + 1$.

b) By sketching $y = \tanh x$ and $y = x + 1$ on the same axes, determine the number of stationary points the curve $y = f(x)$ has.

Question 41. It is given that the curve with equation $y = \dfrac{\sinh x + 1}{\tanh x}$ has a single stationary point. Find the exact coordinates of the stationary point.

Question 42. It is given that the curve with equation $y = \dfrac{\cosh x}{4 - \sinh x}$ has a gradient of $\frac{1}{16}$ at two points. One of these points is $x = 0$. Find the exact x-coordinate of the other point.

Question 43. Show that the curve $y = \dfrac{4}{\cosh x}$ has two points of inflection and find their exact coordinates.
The following questions use the derivatives of $\operatorname{sech} x$, $\operatorname{cosech} x$ **and** $\coth x$ **and use any differentiation knowledge from A-Level Maths.**

Question 44. By writing $\coth x$ as $\dfrac{\cosh x}{\sinh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$.

Question 45. By writing $\operatorname{cosech} x$ as $\dfrac{1}{\sinh x}$, use the quotient rule to show that

$$\frac{d}{dx}(\operatorname{cosech} x) = -\coth x \operatorname{cosech} x.$$

Question 46. By writing $\operatorname{sech} x$ as $\dfrac{1}{\cosh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$.

Question 47. Show that the curve with equation $y = 2x + \operatorname{sech} x$ has two points of inflection. **Prerequisite knowledge for Q48: Hyperbolic Identities**

Question 48. Show that $\dfrac{d}{dx}\left(\ln \tanh \dfrac{x}{2}\right) = \operatorname{cosech} x$. **Prerequisite knowledge for Q49: Differentiation of Inverse Trigonometric Functions**

Question 49. Given that $y = \arctan(4e^{5x})$, show that $\dfrac{dy}{dx} = \dfrac{20}{p\sinh 5x + q\cosh 5x}$ where p and q are integers to be found.

Hyperbolic Functions: Integration

Questions 1–25

Question 1. Find:

a) $\int \sinh x \, dx$

b) $\int \cosh x \, dx$

c) $\int \operatorname{sech}^2 x \, dx$

d) $\int \sinh 3x \, dx$

e) $\int \cosh \frac{x}{2} \, dx$

f) $\int \operatorname{sech}^2 9x \, dx$

Question 2. Evaluate the following definite integral:

$$\int_{\ln 2}^{\ln 3} \sinh x \, dx$$

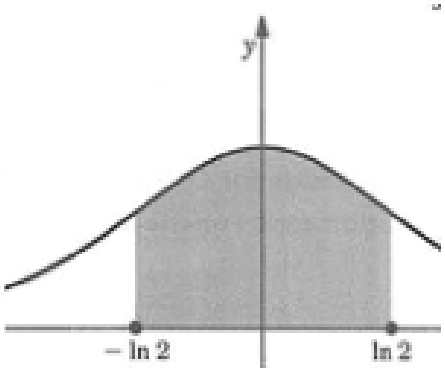
Question 3. Evaluate the following definite integral:

$$\int_{\ln 3}^{\ln 4} \cosh 2x \, dx$$

Question 4. Evaluate the following definite integral:

$$\int_{0}^{\ln 2} (15 \cosh 3x - 9 \sinh 3x) \, dx$$

Question 5. The diagram below shows the curve $y = \text{sech}^2 x$. Show that the exact shaded area is $\frac{6}{5}$.



Question 6. Find the exact value of:

$$\lim_{\delta x \to 0} \sum_{x=0}^{\ln 2} (8\cosh 4x - 5\sinh 3x)\,\delta x$$

Question 7. It is given that $f(x) = 6\sinh 3x + 15\cosh 3x$. Find the exact positive value of k for which the region between the curve $y = f(x)$, the x-axis, the y-axis, and the line $x = k$ is 8. **Prerequisite knowledge for Q8: Integration of Reciprocal Trigonometric Functions**

Question 8. Show that

$$\int_{-\pi/4}^{\pi/4} (\sec^2 x - \operatorname{sech}^2 x)\, dx > \frac{4}{1+e^{\pi}}.$$

Question 9. Find:

a) $\int \sinh(2x+5)\,dx$

b) $\int 8\cosh(3-4x)\,dx$

c) $\int 2x\,\operatorname{sech}^2 x^2\,dx$

d) $\int x\sinh(3x^2-2)\,dx$

e) $\int \dfrac{\cosh\sqrt{x}}{2\sqrt{x}}\,dx$

f) $\int \dfrac{\operatorname{sech}^2(3/x^2)}{4x^3}\,dx$

Question 10. Evaluate the following definite integral:

$$\int_0^{\ln 2} \sinh\left(\frac{3x+\ln 4}{2}\right)\,dx$$

Question 11. Evaluate the following definite integral:

$$\int_0^{\sqrt{\ln 2}} 256x \cosh(4x^2)\, dx$$

Question 12. Show that:

$$\int_0^{\pi/2} \cos x \sinh(\sin x)\, dx = \frac{(e-1)^2}{2e}.$$

Question 13. Find:

a) $\int \tanh x \, dx$

b) $\int \coth x \, dx$

Question 14. Show that

$$\int_{\ln 2}^{\ln 4} (2\tanh 2x + 2\coth 2x)\, dx = \ln\left(\frac{p}{q}\right),$$

where p and q are integers and $\frac{p}{q}$ is in its simplest form.

Question 15.

a) Find $\frac{d}{dx}(\operatorname{cosech} x)$.

b) Hence find $\int \operatorname{cosech} x \coth x \, dx$.

Question 16.

a) Find $\frac{d}{dx}(\operatorname{sech} x)$.

b) Hence find $\int \operatorname{sech} x \tanh x \, dx$.

Question 17.

a) Find $\dfrac{d}{dx}(\coth x)$.

b) Hence find $\displaystyle\int \operatorname{cosech}^2 x\, dx$.

The following questions in this section require the use of Integration by Substitution, Integration by Parts, Hyperbolic Double Angle Formulae and Partial Fractions.

Question 18. Find:

a) $\displaystyle\int x \sinh 3x\, dx$

b) $\displaystyle\int x^2 \cosh \frac{x}{4}\, dx$

Question 19. The curve has gradient function

$$\frac{dy}{dx} = 9(x+2)\sinh 3x$$

and passes through the point $(0, 8)$. Show that y can be written in the form

$$y = (ax+b)\cosh 3x - \sinh 3x + c,$$

where a, b and c are integers.

Question 20. Find:

a) $\displaystyle\int e^{x}\sinh x\,dx$

b) $\displaystyle\int e^{2x}\cosh x\,dx$

c) $\displaystyle\int e^{3x}\sinh 2x\,dx$

d) $\displaystyle\int e^{-4x}\cosh 4x\,dx$

Question 21. Find $\int \cosh^2 x \, dx$.

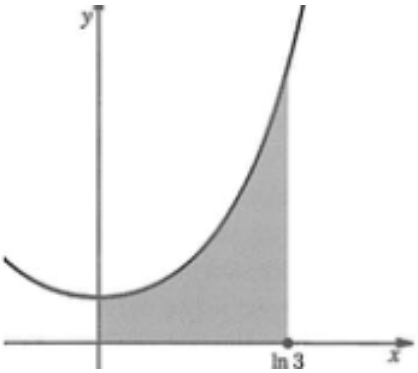
Question 22. Find $\int \sinh^2 6x \, dx$.

Question 23. Show that

$$\int_{\ln(5/6)}^{\ln(6/5)} \cosh^2 \frac{x}{2}\, dx = p + \ln q,$$

where p and q are positive rational numbers.

Question 24. The diagram below shows the curve $y = 4\cosh^2 x - 3\cosh x$. Show that the exact shaded area is $\frac{4}{9} + \ln 9$.



Question 25. Find:

a) $\int \sinh^3 x \, dx$

b) $\int \cosh^3 x \, dx$

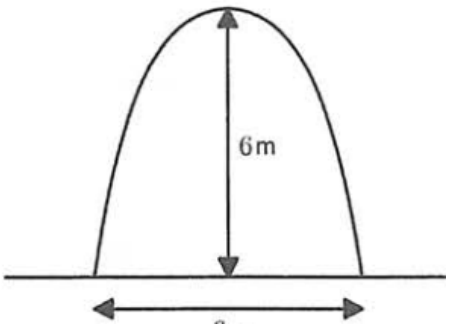
Hyperbolic Functions: Modelling

Questions 1–8

Question 1. The cross-section of a tunnel is modelled by the curve with equation

$$y = 2p - q\cosh x$$

where y is the height, in metres, of the tunnel above the road surface, x is the horizontal distance, in metres, from the centre of the tunnel, and p and q are positive constants. The tunnel is symmetric about the vertical line $x = 0$.



The tunnel is 6 metres wide at road level and has a maximum height of 6 metres.

a) Find the values of p and q to three significant figures.

b) A vehicle of rectangular cross-section is to pass through the tunnel. The vehicle is 4 metres wide and 4 metres high. Determine whether the vehicle can pass through the tunnel without touching the sides.

Prerequisite knowledge for Q2: Mean Value and Reversing the Chain Rule

Question 2. The temperature, $T°\text{C}$, of a chemical solution t hours after a catalyst is added can be modelled by the equation

$$T = 60 - 15\tanh 0.8t$$

for $t \geq 0$. Using the model, determine the mean temperature of the solution during the first 45 minutes of the catalyst being added, giving your answer correct to two decimal places.

Question 3. The velocity, $v\,\mathrm{m\,s^{-1}}$, of a skydiver t seconds after leaving a stationary helicopter is modelled by

$$v = 55\tanh(0.12t)$$

for $t \geq 0$.

a) Explain why this model predicts the existence of a terminal velocity, and state its value.

b) According to the model, find the time taken for the skydiver to reach 90% of the terminal velocity. Give your answer to 2 decimal places.

Prerequisite knowledge for Q4: Volume of Revolution and Hyperbolic Identities

Question 4. A water treatment company is designing a large filtration pipe. The inside wall of the pipe is modelled by the curve

$$y = 2 + \cosh\left(\frac{x}{2}\right)$$

for $-4 \leq x \leq 4$, where x and y are measured in metres. The pipe is formed by rotating the region enclosed by the curve and the x-axis about the x-axis.

a) Show that the volume V of the pipe is 381 m^3 to the nearest cubic metre.

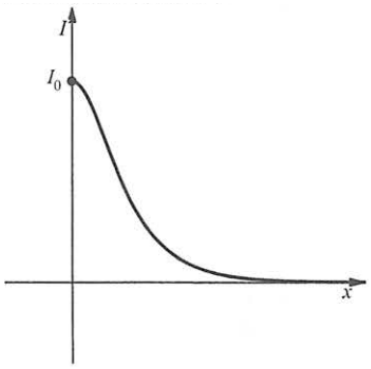
b) The pipe can process 0.85 m^3 of water per second when completely full. Estimate how many seconds are needed to completely fill the pipe.

Prerequisite knowledge for Q5: Differentiation of Hyperbolic Functions

Question 5. Light travelling through water decreases in intensity with depth. In a particular model, the intensity I at a depth of x metres below the surface is given by

$$I = I_0 \operatorname{sech}(kx)$$

where I_0 is the intensity at the surface, k is a positive constant, and $x \geq 0$.



a) Show that the depth x at which the intensity is equal to fI_0, where $0 < f < 1$, is given by $x = \dfrac{1}{k} \operatorname{arcosh}\left(\dfrac{1}{f}\right)$.

b) Given that the intensity is reduced to 40% of its surface value at a depth of 12 metres, find the value of k, giving your answer to 3 significant figures.

c) Given that $I_0 = 120$, find an expression for $\dfrac{dI}{dx}$ in terms of k and x.

d) Using your value of k from part (b), find the rate of change of light intensity with respect to depth when $x = 8$, giving your answer to 3 significant figures.

e) Show that the magnitude of $\dfrac{dI}{dx}$ decreases as x increases for $x \geq 0$. Hence explain what this suggests about the behaviour of light as it enters the water.

f) Given that $\operatorname{sech} u \approx 1 - \frac{u^2}{2}$ for small values of u, comment on how well the model represents light intensity close to the surface.

Question 6.

a) Show that $e^{kt} \equiv \cosh kt + \sinh kt$.

The displacement, x metres, of a particle at time t seconds is governed by the differential equation

$$\frac{d^2x}{dt^2} - 9x = 0.$$

Initially, the particle is at $x = 2$ and is moving with velocity 3 $\mathrm{m\,s^{-1}}$.

b) Show that the general solution can be written in the form $x = A\cosh 3t + B\sinh 3t$, where A and B are constants to be found.

c) Express the solution in the form $x = R\cosh(3t + \alpha)$.

d) Find the time at which the displacement of the particle is $2\sqrt{3}$ metres, giving your answer in the form $\dfrac{1}{6}\ln(a + b\sqrt{3})$, where a and b are rational numbers.

Prerequisite knowledge for Q7: Differentiation and Integration of Hyperbolic Functions

Question 7. A small research rocket is launched vertically upwards. Its velocity, $v\,\mathrm{m\,s^{-1}}$, at time t seconds after launch is modelled by the equation

$$v = A\sinh kt$$

where A and k are positive constants.

a) Show that the acceleration of the rocket, $a\,\mathrm{m\,s^{-2}}$, can be expressed in the form $a = k\sqrt{v^2 + A^2}$.

b) It is given that the initial acceleration of the rocket is $2\ \mathrm{m\,s^{-2}}$. It is also known that when the rocket reaches a velocity of $12\ \mathrm{m\,s^{-1}}$, its acceleration is $5.2\ \mathrm{m\,s^{-2}}$. Find the value of A.

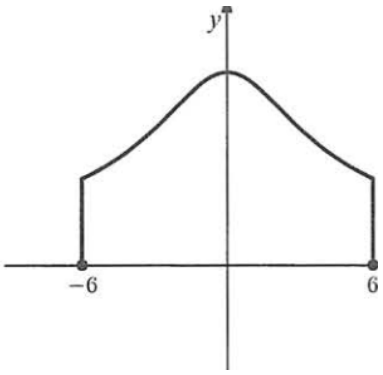
c) Show that $k = 0.4$.

d) Find the distance travelled by the rocket in the first 5 seconds of its launch, giving your answer to the nearest metre.

Prerequisite knowledge for Q8: Integration using Inverse Hyperbolic Functions

Question 8. A farmer is designing a new barn for storing hay. The cross-section of the barn is modelled for $-6 \leq x \leq 6$ by the curve

$$y = \frac{12}{\sqrt{x^2 + 9}}.$$



The region enclosed by the curve, the x-axis, and the lines $x = -6$ and $x = 6$ represents the cross-sectional area of the barn. Each unit on the axes represents 2 metres. The barn is modelled as a prism of length 18 metres.

a) Show that the cross-sectional area of the barn is $96\ln(2 + \sqrt{5})\,\mathrm{m^2}$.

b) Find the volume of the barn, giving your answer to the nearest $\mathrm{m^3}$.

c) The farmer stores hay bales occupying $1.8\ \mathrm{m^3}$. Estimate the maximum number of hay bales that can fit inside the barn.

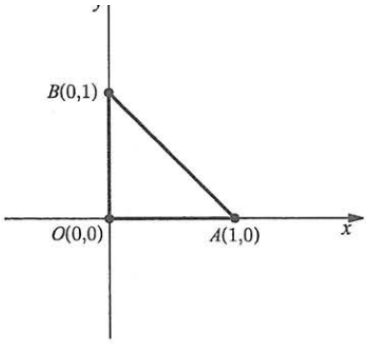
Hyperbolic Functions: Mix

Questions 1–13

Question 1. The transformation T is represented by the matrix $M = \begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix}$, where $x > 0$.

a) Show that T preserves area.

b) Find the exact value of x that maps the triangle shown in the diagram below onto an equilateral triangle.



Prerequisite knowledge for Q2: Volume of Revolution, Hyperbolic Identities

Question 2. It is given that $f(x) = 4\sinh x - 6\cosh x$.

a) Show that the curve $y = f(x)$ has a turning point at $x = \frac{1}{2}\ln 5$ and determine its type.

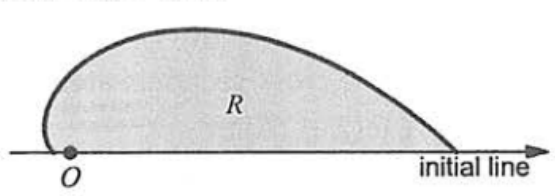
b) The arc of the curve $y = f(x)$ between $x = 0$ and $x = \frac{1}{2}\ln 5$ is rotated through 2π radians about the x-axis. Show that V, the volume generated, is given by $\pi(p + q\ln 5)$, where p and q are integers.

Prerequisite knowledge for Q3: Polar Integration

Question 3. The diagram below shows the curve with polar equation

$$r = 4\sqrt{\cosh 2\theta - \sinh 2\theta},$$

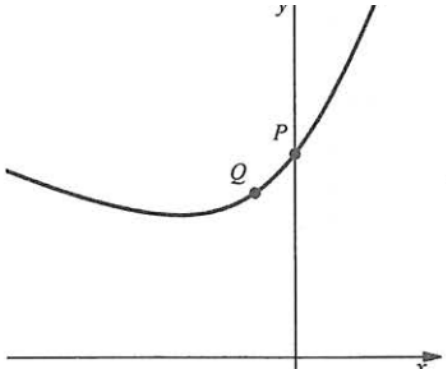
where $0 \leq \theta \leq 2\pi$.



The region R is bounded by the initial line, the curve, and the line with $\theta = \pi$. Find the exact area of R, giving your answer in the form $p + qe^{r\pi}$, where p, q and r are integers. **Prerequisite knowledge for Q4–Q5: Parametric Equations and Hyperbolic Differentiation**

Question 4. The diagram below shows the curve with parametric equations

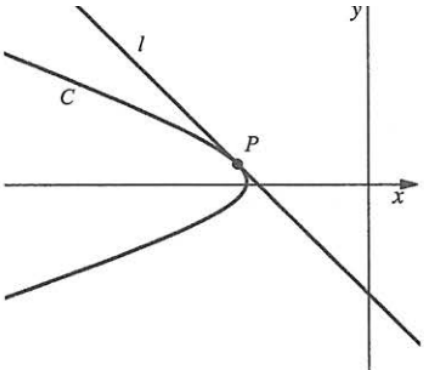
$$x = 3\sinh t - \cosh t \qquad \text{and} \qquad y = \sinh t + 2\cosh t.$$



a) The curve intersects the y-axis at P. Find the exact coordinates of the point P.
b) Q is the point on the curve when $t = 0$. Find the equation of the tangent to the curve at Q in the form $ax+by = c$, where a, b and c are integers.

Question 5. The diagram below shows the curve C with parametric equations

$$x = 5\sinh t - 8\cosh t \qquad \text{and} \qquad y = 2\cosh t - 3\sinh t.$$

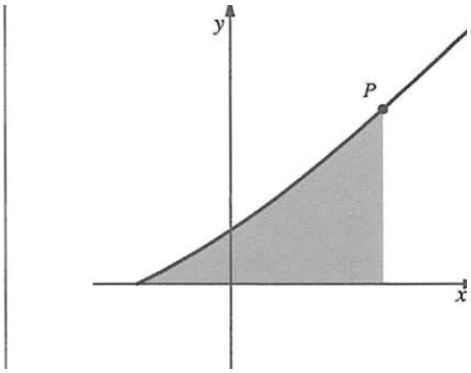


The line l is a tangent to the curve at P, where $x + y$ is at its maximum. Find the equation of the line l in the form $x + y = k$, where k is an exact constant. **Prerequisite knowledge for Q6: Parametric Equations and Hyperbolic Identities**

Question 6. The diagram below shows the curve with parametric equations

$$x = 2\sinh 2t - \cosh 2t, \qquad y = \sinh 2t$$

for $t \geq 0$. It is given that P is the point on the curve with $t = \ln 2$. The shaded region is the region enclosed by the curve, the x-axis, and the vertical line from the x-axis through P.



a) Show that the Cartesian equation of the curve is $x^2 + 3y^2 = 4xy + 1$.
b) Show that the area of the shaded region can be written in the form $a + \ln b$, where a is a rational number in its simplest form and b is a positive integer.

Prerequisite knowledge for Q7: Euler's Formula

Question 7. It is given that z is a complex number.

a) By using Euler's formula, show that $\cosh(iz) = \cos z$.

b) Assuming that the equation has complex roots, find in logarithmic form the roots of the equation $\cos z = 4$.

Prerequisite knowledge for Q8: Integration with Inverse Trigonometric Functions

Question 8.

a) Show that $3\sinh x + 5\cosh x$ can be written in the form $e^{-x}(p + qe^{2x})$, where p and q are integers.

b) Hence, using a suitable substitution, show that

$$\int_{\ln(\sqrt{3}/6)}^{\ln(1/2)} \frac{1}{3\sinh x + 5\cosh x}\, dx = \frac{\pi}{24}.$$

Prerequisite knowledge for Q9: Maclaurin Series

Question 9. It is given that

$$f(x) = \operatorname{artanh}\left(\frac{2x-1}{2-2x}\right), \qquad |x| < \frac{3}{4}.$$

a) Write $f(0)$ in the form $\ln\left(\frac{\sqrt{k}}{k}\right)$, where k is a positive integer.

b) Show that $f'(x) = \frac{2}{3-4x}$.

c) Find $f''(x)$.

d) Hence show that the Maclaurin series for $f(x)$, up to and including the x^2 term, can be written as $\ln\left(\frac{\sqrt{k}}{k}\right) + px + qx^2$, where p and q are constants to be found.

Prerequisite knowledge for Q10: De Moivre's Theorem and Mean Value

Question 10. It is given that $f(x) = \sinh ix$, where $0 \leq x \leq 2\pi$. Find the mean value of $y = (f(x))^4$.

Question 11.

a) Given that $\cosh(A+B) = \cosh A \cosh B + \sinh A \sinh B$, express $\cosh((p+1)x)$ and $\cosh((p-1)x)$ in terms of $\cosh(px)$, $\sinh(px)$, $\cosh x$, and $\sinh x$.

b) Hence show that

$$\sinh(px) = \frac{\cosh((p+1)x) - \cosh((p-1)x)}{2\sinh x}.$$

c) Hence find the sum of the series $S_n = \sinh x + \sinh 2x + \cdots + \sinh nx$ in terms of $\cosh x$, $\sinh x$, $\cosh nx$, and $\cosh((n+1)x)$.

Prerequisite knowledge for Q12–Q13: Integrating Factors for First Order Differential Equations

Question 12. Given that $y = 2$ when $x = \ln 2$, solve the differential equation

$$\frac{dy}{dx} + y\coth x = \cosh^2 x.$$

Question 13. Given that $y = 5$ when $x = 0$, solve the differential equation

$$\frac{dy}{dx} - y \tanh x = \sinh 2x.$$